



MMS – Opportunities, migration and profits
White paper

Version 1.2

MATERNA
Information & Communications

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About MATERNA

The MATERNA group is a leading software distributor specialising in information and communications technology. The company currently employs over 1,400 employees worldwide with a turnover in 2000 of 178 million Euro (DM 349 million). In addition to its headquarters in Dortmund, the company has branch offices throughout the whole of Germany as well as in France, Austria, Belgium, Denmark, the Netherlands, United Kingdom, the Czech Republic, Thailand and Hong Kong. MATERNA's Business Unit Information creates e-business solutions for electronic trading on the Internet. The Internet strategy "e³-solutions" comprises services in the fields of e-business, e-customer relationship management and e-management. The Business Unit Communications offers products and solutions in the areas of mobile solutions and unified messaging. These primarily include mobile value added services based on WAP and SMS which are marketed under the brand name Anny Way.

You can access further information about MATERNA on the Internet at: <http://www.annyway.com>

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Creating the Messaging Future

There is no doubt about it: **Multimedia Messaging** announces a new era of mobile communication and will drive the mobile data services business for 2.5G and 3G networks.

This is good news for you, as a Network Operator: Firstly, new, lucrative business opportunities as a result of this new technology.

Secondly: a unique and total solution is at your disposal world-wide **today**, to enable you to enter the attractive market of Multimedia Messaging **now**.

A major success factor of Multimedia Messaging services is the selection of the optimal business model and the answers to a series of important questions regarding the effective introduction of the new service:

- How can the „Time to Market“ process be accelerated?
- Which migration strategies are available?
- How can MMS be smoothly integrated into existing infrastructures?
- How can Legacy Phones be supported?
- How can successful MMS applications be identified and commercialized?

This strategy paper provides the **answers** to these important questions.

The Commercial Opportunity for MMS

In the past, network operators have been able to achieve a significant increase in usage of mobile data services. Thus, mobile data services make up an estimated 5 to 10% of ARPU (average revenue per user) today: a considerable increase in this percentage is expected in the future.

Messaging is the core of mobile data services – as demonstrated by the success of the Short Message Service (SMS), which has focused the interest of the entire mobile telephony industry.

The introduction of the next Multimedia Messaging generation represents therefore a **huge opportunity** for all members of the value chain - for example increase in market share, increase in ARPU and return on the enormous investments for 3G-network infrastructure or licenses.

MMS Opportunities

- MMS provides a new range of full-multimedia applications and services
- MMS is based on the successful features of SMS and improves and extends those capabilities
- New media formats provide additional Mobile Commerce possibilities
- Differentiation of a service portfolio with MMS
- Multimedia features are seen as one of the key reasons for handset purchases (new and replacement sales)
- Merrill Lynch estimates that 56% of the handset sales in 2002 will be replacement sales- MMS answers the consumer's need to share experiences and emotions
- MMS reflects the fact that human communication is primarily visual
- Users will pay more for rich MMS services than for SMS services
- MMS addresses subscribers who ignore SMS because of complicated text-input
- New and simple communication is possible: e.g. sending a voice-mail to many recipients
- MMS goes hand in hand with the success of digital photography and digital imaging
- MMS may substitute postcards
- MMS is the key to providing 3G services in the future

Fascinating Potential

MMS takes current SMS services to a **new dimension** and offers new, fascinating functionalities in addition to a whole range of new business opportunities. The Multimedia Messaging Service is not only a new service, but a completely new communications medium - which adopts important SMS success factors.

By linking multimedia contents with complementary technologies, such as Location Based Services (LBS), Mobile Commerce will also be driven forward.

The long-term **potential** of the Multimedia Messaging Services is very impressive:

- Non-voice service revenues will exceed voice revenues by 2004 and comprise 66% of 3G service revenues by 2010.
 - In 2010 the average 3G subscriber will spend approximately \$30 per month on 3G data services.
- (Source:UMTS-Forum)

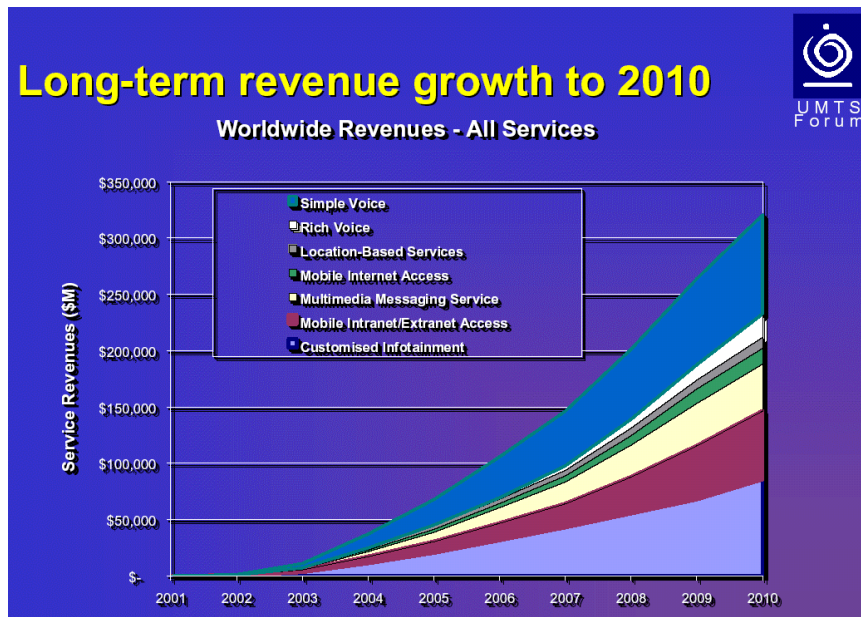


Figure 1 - Worldwide Revenues All Services (Source UMTS Forum)

The Migration

The migration to Multimedia Messaging brings new challenges for the network operator: positioning, interoperability, pricing, billing and the rise of a new generation of experienced users with a new set of requirements.

Why MMS and not Mobile E-Mail will succeed

There is a certain amount of overlap between E-Mail and MMS services. There are, however, some important aspects which prove the superiority of MMS in the mobile arena. The following table provides an overview:

Mobile E-Mail	MMS
- Requires higher attention of the user - Service cannot be adapted to the requirements of mobile users	- Requires low attention of a mobile user - Service can be adapted to the requirements of mobile users
- Attachments must be opened manually	- No need for attachments - One integrated message, single key access to the message
- User must decide in which order he wishes to view content	- Multimedia elements are presented in a synchronized manner - MMS presentation is based upon the synchronization language (SMIL). - Message is experienced as a slide show - The message is one entity
- E-Mail is often a time-delayed medium, as people do not read and respond to E-Mail immediately	- MMS will be immediate and "in the pocket"
- Pull Service - User must actively download E-Mail and does not automatically know when he has received new mail.	- Push Service – user is notified when he has received a new message
- E-Mail Server is required	- E-Mail Server is not required
	- Location Based Services may be combined with MMS
	- Content Conversion in order to ensure optimal display
- Users are not prepared to pay for E-Mail	- Users are prepared to pay for MMS messages
	- Various billing possibilities - Support for reply-charging and prepaid services

Understanding the Customer

In order to understand the opportunities of Multimedia Messaging, the customer requirements must be considered: Who is the customer? What motivates him to use MMS? What expectations does he have? What type of services are required?

As leaders in providing successful solutions for mobile business, Materna will continue to deliver customer orientated applications to the operator at an early stage.

Winning Applications and Services

MMS means a giant leap forward for the mobile industry, creating a revolutionary messaging standard and triggering a dynamic market evolution for consumer data services.



Figure 3 - Building on successful SMS applications

MMS improves almost everything

Applications development will expand far beyond current levels, giving total mobility to multimedia telecommunications. Using MMS, messages can be sent with content as varied and complex as text, graphics, sound, images and videos, all combined into one message, as an MMS message can contain either single or multiple media.

Furthermore, MMS will support new types of media as they gain popular acceptance. The potential applications are numerous. Picture postcards may be sent, video-clips containing, for example, the latest football goals, animated weather maps, daily cartoons and much more.

Winning solutions

Materna offers a complete MMS applications portfolio, with services tailored to specific users and market segments: Value Added Services which keep you abreast with customer requirements and ahead of your competitors, thus supporting your business model to improve on Mobile data service revenue. By satisfying customer demand and automating your customer billing, Materna offers MMS solutions that cater to all your needs.

MMS success factors

Successful MMS applications must respond to the unusual demands of the mobile market. Multimedia

Messaging Service (MMS) facilitates three factors that boost information value for mobile device users:

- **Personalization**
- **Time-sensitivity**
- **Mobility**

Applying these three factors to your products and services will generate new product characteristics providing real value to the customer.

Location Based Services

MMS will enhance the attractiveness of currently available location-based services, such as City on Air. Subscribers will be offered a series of alternative goals by means of graphical navigation, for example sights, restaurants, or other MMS subscribers. Descriptions, photos and videos will let the subscriber conveniently choose one specific goal.

Push Services

Some applications are based on push services, such as the daily weather report. Other applications, such as the storage of messages, may be integrated into an operator portal. The pull mechanism is, of course, also supported, for example the user requests a video of the latest football goals. It is possible to send messages from one mobile station to another and from a mobile station to E-mail and visa versa in both cases as person-to-person or person-to-group messaging.

Content Providing

MMS content providing is a huge opportunity as it fulfills all the criteria of modern media consumption: rapid, personal, audio and visual. MMS can be a springboard for content providers to a mass market of value-added services, acting as a new channel for distributing existing content types or offering new types of content that take advantage of MMS's strengths as a mobile distribution channel.

Mobile Marketing/ Advertising

The MMS enabled mobile device creates a direct link between the advertiser and consumer, combining online marketing messages and the physical world of transactions. Generating an opportunity for immediate consumer reaction, contact, or purchase, combined with excellent measurement and tracking capabilities.

The advantages of the MMS mobile marketing campaigns at one view:

- Increased brand awareness
- High involvement of the marketed product with the label and the campaign
- Sharpened customer focus
- Direct customer contact, immediate response possibility (= 1:1 marketing)
- Improved customer service, increased customer added value

Strengthened competitive position and differentiation

- Focused campaigns with less redundancy

- Continuous improvement of services, products and marketing campaigns due to availability and analysis of response data and creation of customer profiles
- Mobile 1:1 Compared to classical direct mailings, mobile marketing campaigns are substantially more effective
- High customer acceptance of mobile marketing
- Location Based Services extend the mobile marketing options

Service Integration

A wide range of value added services may be added to the Multimedia Messaging Service, enhancing the Network Operator's service portfolio.

As a dedicated Value Added Service interface (MM7), the Anny Way MMSC offers two interface technologies, **Corba** and **XML/SOAP**: Corba for high performance, on-site applications and XML/SOAP for integration of 3rd party Value Added Service providers. Both interfaces offer the same API functionality, including certain features which are unique to the MM7 access, such as reverse charging and transfer of rating information.

Developer Community

Materna provides developers with a **broad** range of services:

- **FAQs**
- **Knowledge base**
- **Development tools**
- **Documents download**

Pricing and Billing

Materna can provide business case calculations on demand, which take market research, person-to-person messaging prices and proven content/application figures into consideration.

Prepaid and postpaid billing

The Anny Way MMSC integrates fully into the operator's existing billing infrastructure, both for prepaid and postpaid billing. Internally, a number of trigger points are available which lead to charging events. In the case of **prepaid billing**, a selection of trigger points launch prepaid interrogation – the transaction is suspended and charging data is transferred to the real time interrogation component, which derives a service charge from the rating engine and then reserves the service charge on the prepaid account. Once the prepaid platform has acknowledged that sufficient credit is available, the charge will be deducted from the account and the MMS transaction is resumed.

Hot billing and **batch billing** are also available. CDR files are exported from the BRL database on a regular basis.

The following billing scenarios are possible:

Person-to-person

- sender pays
- no charges for the receiver, except roaming and reverse charging
- sender pays event charge per message
- sender pays per # of recipient per message or receives discount
- including or excluding traffic volume related charge (e.g. GPRS volume charge)

Event Based Charging

- related to message size
- related to message content type
- combination of the above

Content Based Services

- content partner shared revenue model
- ability to reverse charge enables content partners to provide high value push services
- new opportunities through reverse charging mechanism
- greater possibilities for banner advertisement in multimedia messages

The ratio of revenue shares depends on the followings factors

- message termination (e.g. transport fee)
- money collection
- marketing efforts
- value of the services

Prepaid Systems

The majority of today's mobile customers (approx. 50 – 60%) are prepaid customers. It is assumed that this segment will also play a crucial role in the future. Initially, though, one can expect to find MMS Clients in medium and higher priced devices.

We can learn from the worldwide SMS traffic figures that the inclusion of prepaid customers is a significant factor in dramatically increasing messaging volumes. Young and cost-conscious people form the majority of the prepaid customers.

For this reason, network operators need a prepaid system that allows the customer to profit from the advantages of Multimedia Messaging Services and be constantly informed about their current account status. To ensure this, the prepaid billing system must be extremely reliable, performant and be smoothly integrated into the MMSC.

The Anny Way MMSC offers a flexible solution for integration into an existing prepaid billing environment. A prepaid charging interface is used at dedicated points within the message processing to interrogate subscriber credits, reserve amounts, charge and free amounts.

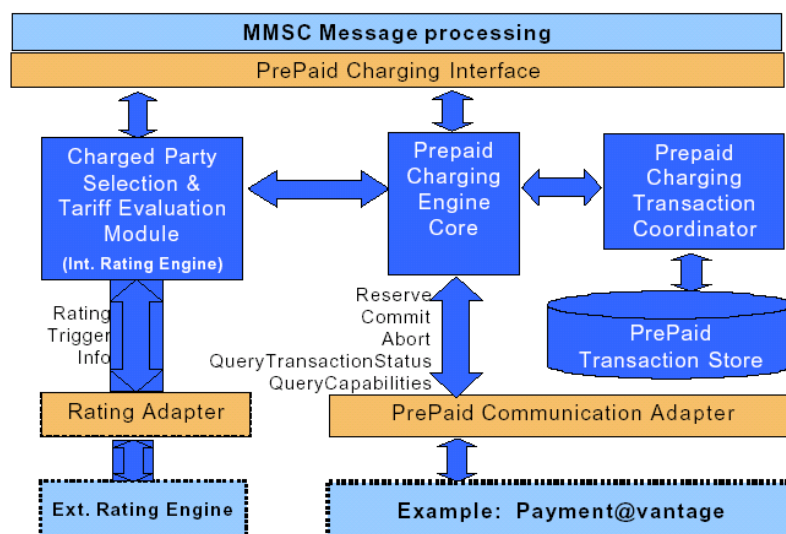


Figure 4 – Prepaid interface

For calculating message costs, an (optional) **rating engine** may be provided by Materna and configured by the operator. The rating engine calculates cost information based on configurable triggers within the message processing. The calculation rules can be configured by the operator. The rating engine performs two functions:

For Mobile Terminated messages, information concerning the cost incurred may be provided with the MM1 notification, informing the subscriber about expected content retrieval costs.

For prepaid customers, cost information may be used for budget interrogation and charging.

Alternatively, billing trigger information can also be provided directly to an existing rating engine.

The prepaid charging engine core offers a generic interface towards external prepaid platforms which can be easily adapted using an integration interface (Prepaid Communication Adapter).

The Prepaid Charging Transaction Coordinator ensures transaction security, particularly if the communication link to the prepaid platform is temporarily broken. Transactions may be stored and finished at a later stage. An operator strategy for this type of fault may be configured.

Message Conversion

Conversion of the message content – or as a minimum parts thereof – is essential to ensure that the recipient is able to view the content with the user agent of his terminal. Terminals may differ widely in terms of display sizes, capabilities and codec types, which makes MMS different from Internet e-mail.

In the Anny Way MMSC, two types of content conversion are supported:

- **Media type conversion** (e.g. Video to a series of images)
- **Media format conversion** (e.g. size reduction, TIFF to JPEG, ...)

Content conversion may take place at two alternative points in the message flow:

- a) **On arrival** of the message at the MMS relay, if the recipient's terminal capabilities are already known.
- b) **After receipt** of the retrieve request (along with the recipient's User Agent Profile).

The content conversion method depends upon the **User Agent Profile** known at stage a) or b), and the recipient's preferences stored in his User Profile. The User Profile settings have precedence over the contents of the User Agent Profile.

Pre-conversion for MM 7 applications. Operators benefit from the Anny Way MMSC's pre-conversion in the form of:

- decreased processing time and MMSC's CPU time
- increased customer satisfaction
- no delay for conversion

Content conversion is registered as a charging event and may be rated for a service surcharge. As content conversion may be a CPU time consuming process in certain cases, it will be handled by separate hardware components in high performance Anny Way MMSC configurations.

Interoperability

Interoperability between different devices and services is of great importance. Today, only the smooth cooperation between all components ensures that services can be developed for devices which will be available on the market in the near future.

Interoperability with all relevant MMS terminals must be an operator's goal.

Benefits:

- High quality end-user service experience
- Enabling MMS as a mass market service

Approach:

- 100% standards compliance
- Comprehensive Interoperability
- Testing (IOT) program
- Handset vendor independence

Integrating Streaming

To meet the high demand for this new area of messaging, Materna integrates a complete streaming solution for Multimedia Messaging. The Anny Way MMSC supports streaming technology as alternative to downloading the MM to the handsets memory. This is very important in case of large Multimedia Messages or insufficient memory capabilities, as the messages would otherwise be discarded.

Depending on operator system environments, two alternatives are possible:

- 1. Integration of a Streaming Server**
- 2. Integration of operator's legacy Streaming Server**

Some content, for example MPEG-4 encoded video material, will not be downloaded to small size terminals, because of limited memory, huge download time or other terminal capability restrictions.

MPEG-4 is the latest international standard issued by the ISO/IEC and developed by its Moving Picture Experts Group (MPEG), the organization that introduced MPEG-1 and MPEG-2. These earlier compression standards serve as the underlying delivery technology for high definition television, digital video broadcasting, direct broadcast by satellite, digital cable television systems, multi-channel-multipoint distribution services, DVDs and interactive CD-ROMs. Building on the success of these earlier standards, MPEG-4 enables integrated, interactive, targeted multimedia delivery across any network - whether broadcast or broadband, wireless or wireline - and via electronic or physical distribution to any device, from set-top boxes to PCs and PDAs.



Figure 5 - Streaming

If the recipient has chosen to receive the message using a streaming mechanism, the streamable multimedia part of the message will be diverted to an external streaming server. The server delivers a reference to the content (e.g. a URL for IP based retrieval or an MSISDN for H.324m or similar access methods), which is used as a replacement for the original multimedia part when delivering the original message. The User Agent of the recipient uses the streaming reference to connect to the streaming engine and view the streaming content. The following diagram shows the concept to integrate streaming components into the MMSC environment:

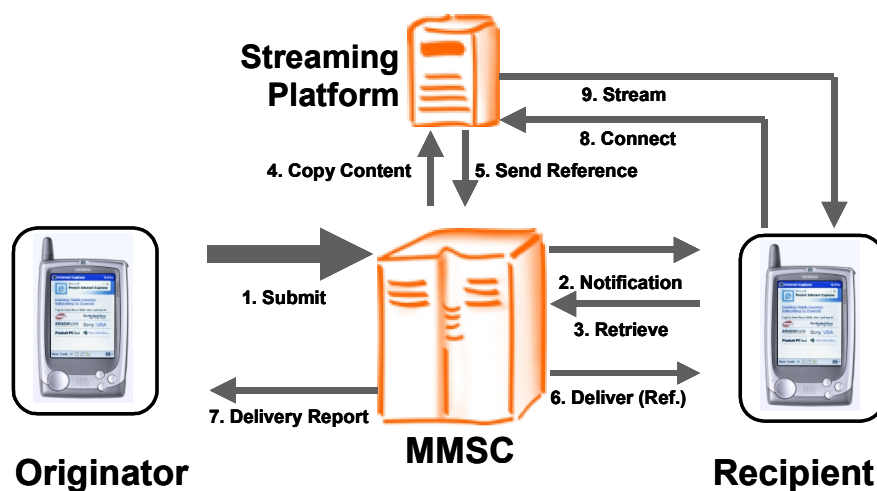


Figure 6 – Streaming Concept

Launching MMS Today

The availability and penetration rate of MMS terminals will be a crucial factor during the roll out of new Messaging Services. The Anny Way MMSC supports all market-relevant MMS terminals of the respective manufacturers.

Roadmap MMS Clients

	Pocket PC with Anny Way MMS Client	Nokia 7650	Ericsson T68
Availability	now	Q2 2002 at the earliest	Q2 2002 at the earliest

As an innovative market leader, Materna brings the Multimedia Messaging Service to the millions of legacy terminals thereby increasing the acceptance of this new service. The MMS Client, for instance, can be installed by the owners of existing **pocket PC** devices.

The Multimedia Messaging Service already has a wide customer acceptance.. Network operators should position themselves today, to be ready for the expected massive market demand .

In addition to the extensive **Legacy Support** offered by Anny Way MMSC, Materna offers the prerequisites for a fast migration of users to attractive MMS services. This eases the purchase of new MMS-enabled terminals and strengthens the customer's relationship with the mobile phone operator.

With the Anny Way MMS **Trial System**, network carriers can today gain experience with Multimedia Messaging services and ensure that they are ideally positioned to address the mass market.

The Pocket PC as alternative

Pocket PC, the Microsoft software for portable pocket computers will have a promising future. The previous version of pocket PC 2002 is already witnessing a great success in sales to end customers all around the world. In April 2001 already, one year after market launch, Microsoft hardware partners reported sales of one million pocket PCs based on Windows CE. According to a study carried out by the English market research company Canalys.com, the market share of pocket PC rose, in the case of Handheld devices and in Western Europe alone, from 13,5 % in the second quarter 2000 to 39,8 % in the second quarter 2001.

Materna's **MMS Client** supports both versions of the successful operating system and constitutes thereby a good basis for an early launch of the Multimedia Messaging Services.

The MMS Client as Innovator

The Multimedia Messaging era can begin immediately: With the support of Pocket PC, Materna already supplies **today** the network operators with a pioneering MMS enabled telephone.

Customizing the MMS Client

Due to the unique customizing possibilities of the MMS Client, network operators can strengthen the customer relationship or build up new sale and cooperation channels. MMS Client customizing may consist of:

- Adaptation to the **Corporate Design** of the network carrier by means of Skin technology
- Unified Messaging
- Automatic preconfiguration
- Integration of **Push Channels**/News Tickers
- **Mobile Marketing/Advertising** Services

Final result: Your „personal“ MMS Client allows you to differentiate from your competitors and position yourself as an innovator in the market.

Testing MMS now with a Trial System

The MMS Trial system - consisting of MMSC, MMS Client, MMS services and MMS-enabled WAP Push/Pull Gateway makes extensive field tests possible. In addition, the mobile communication operator can collect data on customer acceptance and define its non-standard focus. Materna offers both installation of the MMS system at the customer premises as well as Hosting in the Materna data processing center. In the case of MMS Hosting operation, Materna establishes a connection with the operator's GPRS network and the SMSC of the customer providing an authentic solution.

As a part of the Trial version, the Anny Way Picture Postcard application and customized extensions, such as a **Unified Messaging** integration, can be provided on request.

Seamless Integration into Existing Environments

MMSC must be integrated into an existing network environment including a number of legacy systems. In addition, there are various customer strategies regarding an MMS integration. Therefore the Anny Way MMSC design follows a modular and flexible approach including an implementation of abstraction layers which can be easily adapted to different 3rd party systems.

Legacy Support

MMS will take off only when most people have MMS-capable handsets. For this reason, the expected delay in the mass-market availability of dedicated MMS handset may slow down the MMS starting schedule.

Providing operators with a significant market advantage

By giving customers with legacy devices a taste of Multimedia Messaging, this problem can be solved and the roll out of MMS may be accelerated.

Anny Way MMSC provides message storage and support for non-multimedia terminals. The delivery of multimedia messaging is guaranteed regardless of the terminal capability.

Legacy devices can access MMS messages via adapters to widely accepted clients - mobile and PC Web browsers, WAP browsers and E-Mail clients.

Mail Store Integration

In addition to the functionality of temporarily storing Multimedia Messages in the high-speed message Repository of the MMSC when recipients are not available (which is mandatory), the Anny Way MMSC supports the server based inbox concept. This means integration of mail store systems to allow e.g. copying or diverting of messages towards user mailboxes.

Users can access their MMS messages stored in this Mailstore with a normal E-Mail Client or a WEB/WAP client and do not need to worry about expiry times. In order to support different operator strategies and system environments, the following three system models are supported:

- 1. Mail Store integration within the Anny Way MMSC**
- 2. Mail Store integration within operator environment (on project basis)**
- 3. Integration of user dependant external mail stores (like Hotmail, AOL, Yahoo etc.)**

Offering the Perfect Solution

Materna is convinced that Multimedia Messaging will be a key revenue generator for GSM/GPRS/and UMTS networks, continuing the success of SMS by extending the proven SMS business models with a rich user experience.

The unbeatable solution

Profitable partnerships will be a key factor in securing an early lead in this new area of telecommunications. Materna makes your first logical choice and offers a **complete, end-to-end MMS solution** that has been developed on Best-of-Breed philosophy. The comprehensive turnkey service solution includes Multimedia Messaging Service Center, MMS Client, multimedia applications, Value-Added-Service Platform and Integration, comprehensive billing/payment framework, streaming services, and – last but not least – unified multimedia messaging environment.

A common approach

Network operators will benefit from the combination of long-term infrastructure and integration experience and leading mobile Internet Solutions from Materna's high-performance MMSC. In recent years, Materna has established itself as worldwide leader for SMS services. Materna's experience in the creation of successful Value Added Services and network elements (e.g. Anny Way WAP Gateway) is now transferred into a comprehensive MMS offering to mobile network operators.

Advantages for the operator

This unique combination will ensure market success for our customers:

- Complete end-to-end solutions
- Excellent expertise in messaging area
- Worldwide full-service support
- Capability to integrate MMS into any existing infrastructure providing excellent investment security
- MMS Client with SMIL support
- Legacy device support for SMS/WAP terminals
- Comprehensive interoperability, scalability and performance tests
- MMS Service Platform with XML/SOAP support
- Content and Format Conversion Framework (incl. Pre-Conversion Module)
- Powerful billing module with programmable Rating Engine, incl. Prepaid Billing
- MMS Service Platform (VASP Application Server)
- Supplied WAP Browser, WAP Gateway, WAP Push Gateway
- Supplied SMS Service Platform (AIC)
- Mail Store integration: Bundling with Mail Store or integration in existing (e.g. iPlanet)
- active collaboration in 3GPP, WAP Forum and ETSI standardization

Early market entry

The highly scalable MMS architecture in combination with a comprehensive legacy support and the innovative MMS Client allows early market entry whilst keeping initial investments low. It will grow and scale with increasing subscriber numbers, even in large networks. Since it supports different billing models for prepaid and postpaid subscribers, providers have every freedom for designing their individual multimedia messaging business models. In this regard, operators and providers can also benefit from Materna's wide-ranging experience in the area of mobile payment.

Opening up a world of opportunities

Materna Multimedia Messaging Services allows operators to offer their mobile users a rich, compelling and varied user experience today. The MMSC service works over 2/2.5G and 3G networks and provides support of existing and future handsets and PDAs. Materna's MMSC strengthens your market position, enabling you to exploit the opportunities of GSM/ GPRS networks and initiate an early revenue stream - before migrating seamlessly to 3G services.

Worldwide Full-Service

A unique international service portfolio puts our worldwide organization to work for your business, offering you expert consulting, joint marketing, flexible financing and ready-to-use products. Putting you in the position to let your business grow - profitable and with mobility.